

The Effects of Primary School Expenditure on US Student Standardized Testing Outcomes

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ABSTRACT

Inspired by the overwhelming importance of the standardized test in both the development of a primary school student and in the life outcomes of students post-graduation, this poster researched whether or not a correlation exists between primary school expenditures and standardized test results. In order to do so, an extensive literature review of previous educational system research (published on JSTOR, Google Scholar, and NJIT’s Robert Van Houten Database) was conducted for the purpose of detecting aligning and contrasting information regarding standardized testing and school expenditures; this literature review spanned over both the direct question of whether a relationship between school spending and standardized test performance existed as well as the underlying factors embedded within “school expenditure” that could influence this relationship. This poster’s literature review revealed a common thread of small positive associations between school spending and standardized test results, prompting the conclusion that school expenditures indeed do positively correlate with standardized test performance. Further meta-analysis demonstrated that instructional-related spending had the largest positive correlation with standardized test results, whereas other school expenditures (such as infrastructural-related and personnel costs) did not have nearly as strong of a positive relationship in comparison. The discussion section of this poster further develops the logic that enabled these conclusions to be drawn and offers commentary on the social implications of said conclusions. This research poster then concludes with actionable recommendations that school districts may implement to further promote student achievement and proficiency. Keywords: Standardized Test, Spending, Funding, School Expenditures, Test Performance, School Resources, Primary School Standardized Testing, S.A.T., A.C.T., S.T.A.R.R.

OBJECTIVES

- The importance of the standardized test as a mainstay in the educational life cycle of a primary school student has always been heavily contested.
- Prestigious universities are now opting for test-optional policies at an unprecedented rate, prompting discussion on the merits of the standardized test as an accurate metric of success.
- School expenditure, in particular, has been endlessly researched over the past two decades, yet extreme discrepancies between research studies render its effects on standardized testing results ambiguous.
 - An underlying factor behind “school expenditure” exists, causing an indefinite rift between different studies.
 - Further analysis into these discrepancies would reveal if school spending has an association with standardized test performance, and just how strong said association really is (if it exists).

This research poster’s two main research objectives:

- Accurately analyze the correlation between school expenditure and standardized testing outcomes for primary school students
- Evaluate what is causing the preexisting deviation between existing articles covering the topic.

Research Question: *Does a correlation exist between primary school spending and standardized test scoring, and, if so, to what degree?*

MATERIALS & METHODS

In order to find research pertinent to this research paper’s objective:

- JSOR, Google Scholar, and NJIT’s Robert W. Van Houten Library Database were indexed.
- The keywords “standardized test” were interchangeably mixed with “spending” and “funding” when searching the above databases.
- Inputting “school funding standardized,” “school resources standardized testing,” and “school spending test” into the NJIT Robert W. Van Houten Library Database yielded the first three cited sources (as presently ordered in the References section), and the later query of “standardized test expenditure” into Google Scholar and JSTOR yielded the last three sources.
- This paper’s specific research question targets middle and high school students’ standardized testing, meaning that any given cited research resource in this poster utilizes at least one as a research metric.

RESULTS

Analysis of the overall data in the literature review indicates that there exists a positive relationship between general school expenditure and student performance on standardized tests.

The variation of the collected data allows multiple conclusions regarding specific school expenditures to be drawn:

- Instructional-related expenditures hold a large and positive statistically significant relationship with student standardized test performance.
- Infrastructural expenditures, such as upgrading dated teaching equipment, hiring new teachers, and replacing worn school supplies, surprisingly have less of a positive effect on student standardized test achievement than instructional-related expenditures.

Most studies referenced also found that the association between school expenditures and standardized test results was exacerbated slightly in high school tests when compared to middle school tests.

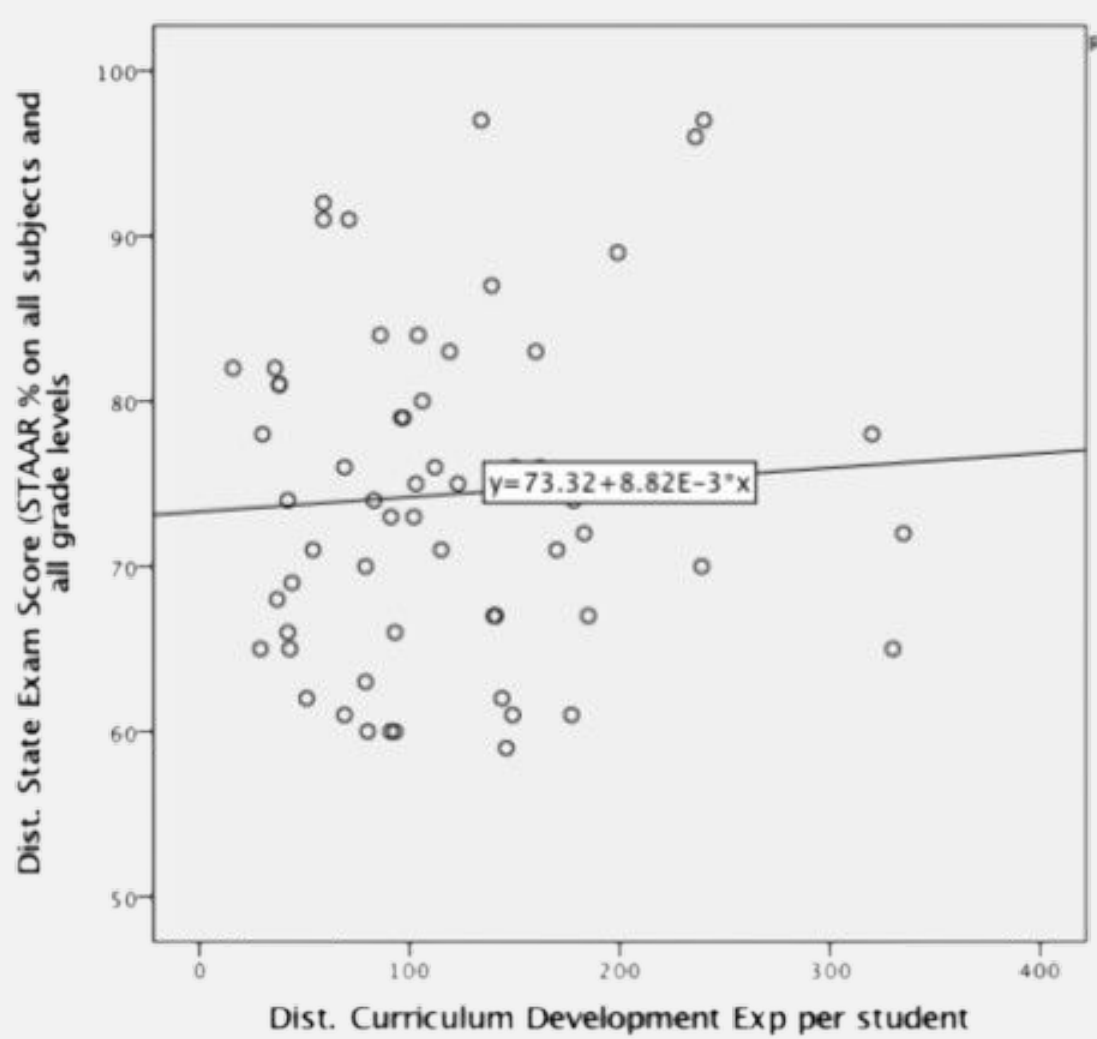


Figure 1. Scatterplot measuring the correlation between Texas STAAR testing scores and per-pupil District Curriculum Development Expenditures (Source: Nicks, Martin, Thibodeau, Young, 2018, p 70.)

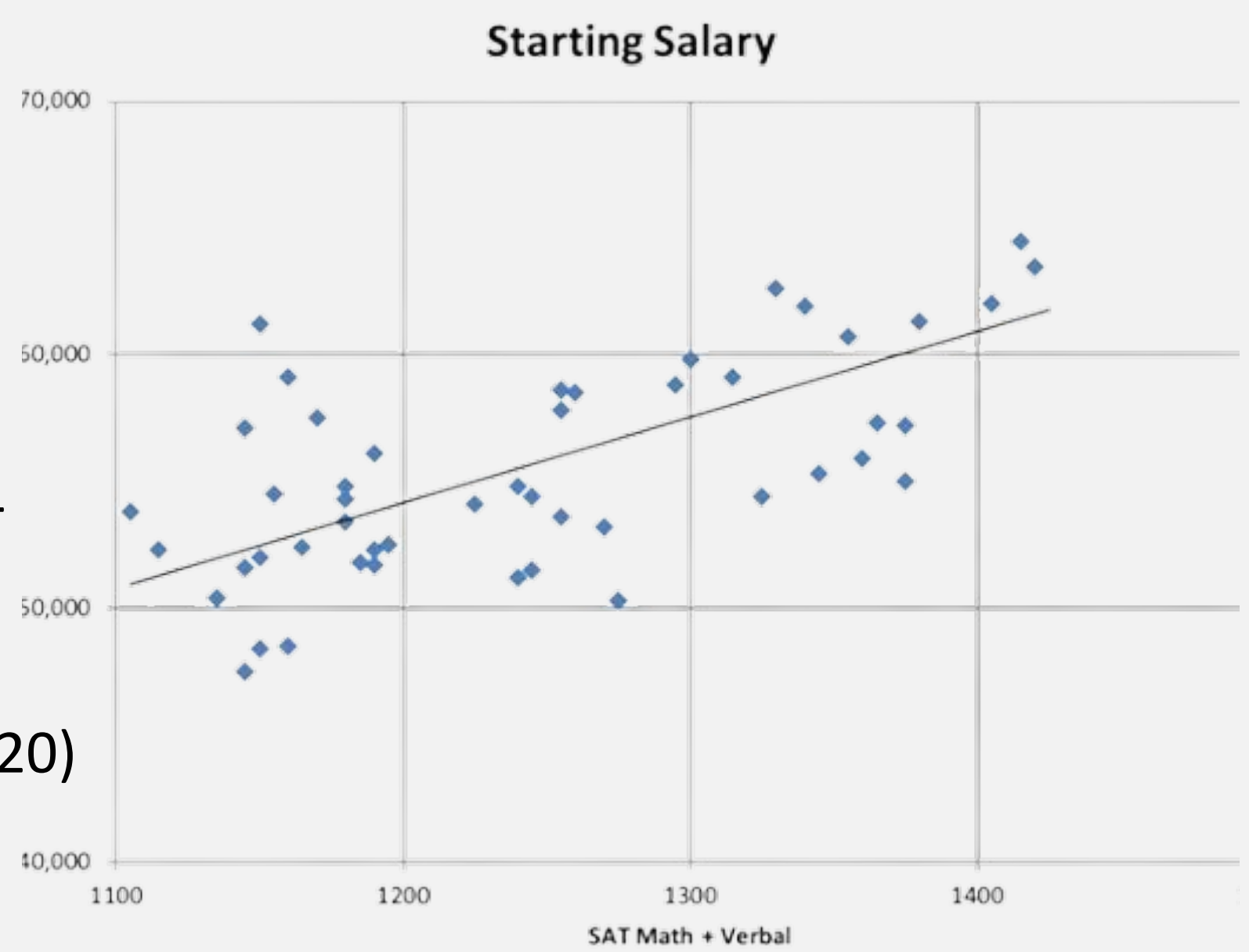
CONCLUSIONS

The claim of a practical and significant positive relationship between instructional-related school expenditures and standardized test accomplishment is a theme throughout the literature due to its direct impact on student learning.

- While other spendings target subfactors of schooling that may only indirectly affect a child’s education, targeted spending, such as on decreasing class sizes, serve to directly influence a student’s learning capabilities, propping each student up as an esteemed member of a classroom rather than one of many students.

Figure 2.

Scatterplot illustrating the positive correlation between composite SAT scores and starting salary (Pinizzotto, 2020)



This paper’s conclusion is, in part, a synthesis of two different fields of thought put forward by existing research:

- One field of thought is that there exists a small, but statistically significant, positive association between the two variables.
- The other field of thought is that the positive link between the two variables is too small to be further generalized.

As indicated by the meta-analysis conducted in this poster, specifically targeting instructional-related expenditures (as previously defined as expenditures pertaining to the direct learning of a student) yields the highest positive interconnection with standardized testing.

- Higher per-pupil district instructional expenditures, smaller class sizes, and after-school tutoring programs all are proven to have positive, statistically-significant correlations with standardized test performance, and all of these factors categorically fall under instructional expenses.

However, infrastructural spending is still integral in keeping a school streamlined and in maintaining productive students.

- With this in mind, one source’s (Sharp) argument that infrastructure-related spending holds a *negative* correlation to test results warrants further discussion.

CONCLUSIONS (CONT.)

Given the importance of the standardized test and the prestige of standardized tests overall, the question of how a given school allocates funds is essential to answer for each and every individual student; this study answered the question of school expenditure’s effect on standardized testing to the best of its ability, concluding that school expenditure did indeed have a slight positive relationship with standardized test performance. Specifically, expenditures aimed toward instructional-related costs were found to yield the highest correlation with standardized test scores. The small positive correlation presented by this study between school expenditures and standardized test results may seem insignificant, but any influence on the keystone standardized test has profound repercussions for primary school students, especially considering the influence of the standardized test on career outcomes. The research question answered in this study will be extremely important in determining the future of the US education system, but the social implications of the standardized test may be even more crucial presently, as the entire purpose of the standardized test itself is being called into question. Regardless of whether or not more prestigious universities continue to do away with the standardized test, the standardized test will remain a spearhead of educational prestige in America, and school expenditures will continue to be key determinants of the ever-critical standardized test.

FUTURE RESEARCH

The conclusion that Sharp drew about there being a negative correlation between school expenditures and standardized test scores was surprising, and their explanation that infrastructure-related expenses have a negative effect on this correlation has massive implications for education overall if accurate.

- Factors such as socioeconomic status, race, and geography all affect standardized test performance to at least some extent, yet remain unaccounted for in the data collected.
- The small nature of the overall relationship between school expenditures and standardized test results is also fairly concerning, as analysis of the data from another perspective may lead to the conclusion that no such relationship exists between the two variables.

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